

Comprehensive Data Analysis & Insight Summary Report

Project: Apple Inc. (AAPL) Stock Market Time-Series Analysis & Feature Engineering

Program: Data Analytics Internship | AnalystLab Africa

1. Dataset Overview

This study investigates the historical financial performance of **Apple Inc. (AAPL)**. The dataset was extracted programmatically using the Python `yfinance` library over a multi-year period.

- **Core Features:** Date (Index), Open Price, High Price, Low Price, Close Price, and Trading Volume.
- **Data Structure:** Sequential time-series records capturing daily market fluctuations, price spreads, and liquidity metrics.

2. Data Transformation & Preprocessing Techniques Applied

- **MultiIndex Flattening:** Addressed hierarchical column structures generated by financial APIs by executing level extraction to normalize column labels for seamless Pandas querying.
- **Datetime Formatting & Indexing:** Converted standard text-based timestamps into robust Pandas `DatetimeIndex` objects to enable precision slicing and time-based resampling.
- **Missing Value Management:** Evaluated and handled initial NaN values introduced during rolling statistical window operations without compromising data integrity.
- **Data Resampling & Grouping:** Aggregated daily records into monthly periodic summaries using Pandas `resample` and `group` operations to observe macro trends.

3. Features Created (Feature Engineering)

- **Daily Price Change:** Absolute monetary difference between daily Close and Open prices ($\text{Close} - \text{Open}$).
- **Percentage Price Change (Daily Returns):** Normalized percentage shifts tracking day-over-day financial growth or decline.
- **Moving Averages (MA):** Calculated short-term (7-day) and medium-term (30-day) rolling means to smooth volatility and highlight directional trends.
- **Rolling Volatility:** Computed the 30-day rolling standard deviation of daily returns to quantify risk and market uncertainty.
- **Intraday High-Low Spread:** Measured daily trading bandwidth ($\text{High} - \text{Low}$) to gauge intra-session volatility.

4. Trends Identified & Key Insights from Analysis

- **Macro Growth Patterns:** The closing price trend exhibits long-term appreciation punctuated by periodic consolidation phases.
- **Volume-Price Correlation:** Significant upward price movements consistently align with massive spikes in Trading Volume, confirming strong institutional participation during breakouts.
- **Moving Average Crossovers:** The 7-day MA reacts rapidly to short-term momentum shifts, whereas the 30-day MA functions as a solid baseline filter, eliminating daily noise and revealing structural trend lines.
- **Volatility Clustering:** Higher volatility periods correspond closely with steep corrections, followed by stabilizing sideways movement.

5. Recommendations Based on Findings

- **Trading Signal Generation:** Implement Moving Average crossovers (7-day intersecting 30-day MA) as an algorithmic trigger for identifying entry and exit points.
- **Risk Management:** Utilize the 30-day rolling volatility metric to dynamically adjust stop-loss thresholds during turbulent market cycles.
- **Liquidity Screening:** Factor in volume trend thresholds before executing capital allocation strategies to prevent false breakout traps.